Yu Miao

347-813-1812 | yumiao0557@gmail.com | yumiao0557.github.io | LinkedIn: https://www.linkedin.com/in/yu-miao-96130a172/

EDUCATION

Duke University **Aug. 2021 – Dec. 2022**
Master of Science in Biomedical Engineering *Durham, NC*

University of California, San Diego **Sept. 2016 – June 2020**
Bachelor of Science in Bioengineering: Biosystems *La Jolla, CA*

SKILLS

Languages: Python, MATLAB
Machine Learning: Machine Learning, Deep Learning frameworks (PyTorch, TensorFlow), LLM-based tooling
Camera & Imaging: ISP Tuning, IQ Evaluation, Noise Characterization, **MTF**/Optical Metrology, **HDR** Processing, Sensor Calibration
Tools & Methods: OpenCV, **IQA**-PyTorch, **Lab Automation**, RAW Image Pipelines, **Camera Test Automation**

PROFESSIONAL EXPERIENCE

Glass Imaging **July 2026 – Present**
Camera Software Engineer - AI Lab Automation, *San Mateo, CA*

- Design **ML based Neural ISP** to replace current traditional ISP. (Joint ML - Demosiac and Denoise)
- Collaborated cross-functionally with ML and hardware teams to define **image quality matrices** such as **MTF** and **SNR**.
- Built an **automated camera capturing platform** through adb tunneling used by the ML team to streamline large-scale image dataset collection across varied lighting, scene, and sensor conditions.
- Designed multi-camera synchronization and capture control software to support high-throughput, repeatable data acquisition workflows for model training and evaluation.

Qualcomm - Camera Team **Jan. 2023 – July 2026**
Camera System Engineer - ISP Team *San Diego, CA*

- IQ Testing & Validation: Performed rigorous **subjective** and **objective** image quality validation (with IQA-PyTorch) on simulated and real captures, applying **SNR-based** noise modeling, **PSNR** and **SSIM** for distortion and structural fidelity, and **MTF** analysis to quantify sharpness-noise trade-offs at the camera system level
- IQ Testing & Validation: Built and curated high-quality RAW Bayer datasets for camera and validation, covering **ultra-low-light**, **HDR** scenes, night scenes, and social-media video use-cases, ensuring statistical coverage
- ISP Simulation & Tuning: Simulated and tuned core ISP stages including demosaicing, **AWB**, **AE**, **AF**, **tone mapping**, **gamma**, **chromatic aberration correction**, **denoising**, and **sharpening** within a RAW-to-YUV pipeline
- Metrology & Calibration: Measured full **MTF** curves for the Sony IMX766 sensor using ISO 12233 slanted-edge methodology, capturing 5+ field positions across 4 rotations with **TE42** Chart. Implemented custom Python+OpenCV analysis to extract spatial frequency response and evaluate optical performance across ultra-low-light to bright daylight conditions. This data informed calibration strategies for **HDR**, autofocus sharpness, and noise reduction module tuning
- Algorithm: Algorithm optimization for **LSC** table and increased the speed of bicubic interpolation by up to 4 times
- Algorithm: Developed and integrated **HDR** motion modeling and evaluation into the camera simulation framework with Python, performing motion threshold testing on dynamic scenes to characterize HDR artifacts and ghosting behavior, and enabling automatic tuning of **HDR** parameters based on simulated motion sensitivity
- ML Algorithm: Implemented ML-based auto-tuning pipeline (IQ Bot) with Depict-QA LLM approach. Teaching LLM to give aesthetic and statistical interpretation of the image and auto-tune it with an internal simulation tool
- ML Algorithm: Developed a ML-based image detection tool to automatically detect TE42 resolution charts' angular shifts for accurate and repeatable MTF measurement (>98% accuracy) for camera tuning and simulation workflows

PUBLICATIONS

Tang, W., **Miao, Y.**, Chen, X, (2026)

- Scale-Dependent Performance of YOLOv12 andYOLOv11 for Automated Blood Cell Detection: A Controlled Benchmark on BCCD

Chikwetu, L., **Miao, Y.**, Woldetensae, M., Bell, D., Goldenholz, D., Dunn, J. (2023)

- "Does deidentification of data from wearable devices give us a false sense of security? A systematic review". The Lancet Digital Health, April 2023.

Luo, W., **Miao, Y.**, Magda, S., Ulg, A. M., Melton, R., Haxton, R. K., Le, L., Airriess, C. (2020)

- "Subject classification for autism spectrum disorder using machine learning and radiomics". Society for Imaging Informatics in Medicine, August 2020.